Modeling of the experimental part

3.1 Goal of the experiments

We want to see if our modified PID controller keeps the water level close to the set point and moves the valve less than a normal PID, when the inflow and outflow are unknown. We also want to know what each extra part of the controller does.

3.2 The simulated process

The process is a tank with a valve. We use a simple model of the water balance, and it is the model of the class TankSimulator.

The variables are:

- PV(k): the water level at step k, in percent of the tank (from 0 to 100).
- SP(k): the set point, the level we want.
- u(k): the valve position, from 0 to 100.
- u0(k): the null position of the valve, where the inflow is equal to the outflow, so the level does not change. The controller does not know it. In S1 it is 50 all the time, and in other situations it can change.
- d(k): a disturbance added to the level, like rain (positive) or evaporation (negative). It is 0 in all the situations, except S4.
- r(k): the gain of the valve at step k, a random number.

We get the model from the water balance. The volume of water changes with the net flow, $A * \frac{dh}{dt} = Q_{in} - Q_{out}$, where A is the area of the tank and h is the level. The valve controls the inflow, $Q_{in} = c * u$, and the outflow Q_{out} is unknown. The null position is $u_0 = \frac{Q_{out}}{c}$, so the net flow is $c * (u - u_0)$. For a step of time Δt, the level changes with $(c * \frac{\Delta t}{A}) * (u - u_0)$. We call $r = c * \frac{\Delta t}{A}$ and we get the equation of the program. It is an integrator model, like the integrator models used for canal pools in [1] and [15].

$$PV(k+1) = PV(k) + r(k) * (u(k) - u_0(k)) + d(k)$$

The unknown inflow and outflow are inside u_0 and r. When they change, u_0 changes. What we know about the process:

- The level is constant only when $u(k) = u_0(k)$ and $d(k) = 0$. So the controller has to find u_0, and it does not know it.
- $r(k)$ is a whole number from 5 to 39, divided by 100, new at every step. It has 35 possible values with the same chance, so its mean is 0.22 and its standard deviation is about 0.10.
- In one step the level changes with maximum 0.39 · max(u_0, 100 − u_0), which is 19.5 for u_0= 50, plus the disturbance.

- The valve moves maximum 3% in one step: $|u(k) - u(k-1)| \leq 3$.
- If the new level would be outside 0 to 100, the program ignores that update and keeps the old level.
- The process runs on a timer, one step every 150 ms, and we count time in steps.
- The program starts with $PV(0) = 10$, $u(0) = 50$ and SP = 74.

We assume that the flow through the valve is linear around u_0, that $r(k)$ is independent from one step to the next, that the sensor reads the level exactly (we add noise only in situation S5), and that the controller makes one decision for each step of the process.